

# Machine Learning — Complete Study Notes

Cleaned, corrected, and reorganized from the provided class transcripts

**Purpose:** These notes preserve the ideas taught in class while correcting important technical inaccuracies and organizing the material into a logical learning sequence.

# 1. Why Machine Learning?

Traditional manual analysis works for small datasets, but it becomes slow and impractical when data grows to thousands, millions, or billions of records. Machine learning allows a computer to learn patterns from historical data and apply those learned patterns to new data automatically.

A machine-learning system does not literally “think like a human.” More accurately, it learns a mathematical relationship or decision rule from examples. Once trained, the model can repeatedly perform predictions or pattern recognition at a scale that would be difficult to handle manually.

**Typical workflow:** collect data → clean and preprocess data → select features → train a model → evaluate it → improve it → use it for predictions.

## 2. Main Types of Machine Learning

**Supervised learning:** The training data contains inputs and known target labels/values. The model learns a mapping from X (features) to y (target). Examples include linear regression, logistic regression, decision trees, random forests, and boosting models.

**Unsupervised learning:** The data does not contain a known target label. The algorithm searches for structure or patterns. Common examples include K-Means, hierarchical clustering, and DBSCAN.

**Semi-supervised learning:** A smaller portion of the data is labeled while a larger portion is unlabeled. The labeled examples help the model make use of the unlabeled data. This is useful when labeling is expensive.

**Reinforcement learning:** An agent interacts with an environment and learns a policy from rewards or penalties. The core elements are the agent, environment, state, action, reward, and policy. It is not simply supervised learning with a correct/incorrect label.

## 3. Supervised Learning: Regression vs Classification

**Regression** predicts a continuous numerical value. Examples: house price, temperature, sales amount, demand, weight, or delivery time.

**Classification** predicts a category or class. Examples: spam/not spam, fraud/not fraud, disease/no disease, or one category among several possible categories.

**Binary classification:** exactly two classes, such as 0 and 1.

**Multiclass classification:** one class is selected from more than two mutually exclusive classes, such as cat, dog, or bird.

**Multilabel classification:** one observation can have multiple labels at the same time. For example, one image may contain both a dog and a cat.

**Important correction:** multiclass and multilabel are often confused. “One image containing multiple objects” is multilabel if multiple labels are assigned simultaneously; multiclass means choosing one class from several alternatives.

## 4. Linear Regression

Linear regression models the relationship between one or more input features and a continuous target. In the simplest case, the relationship is represented by a straight line.

**Simple linear equation:**  $y = mx + c$ , where  $x$  is the input,  $y$  is the predicted output,  $m$  is the slope, and  $c$  is the intercept.

The model tries to find parameters that make predictions as close as possible to the actual values. The difference between actual and predicted values is called a residual or error.

**Common regression metrics:** MAE measures average absolute error; MSE squares the errors and penalizes large errors more strongly; RMSE is the square root of MSE and is in the target's original units;  $R^2$  measures the proportion of variance explained by the model.

**Best-fit line:** The goal is not simply to draw a line through the maximum number of points. The model chooses coefficients that minimize a defined loss function, commonly the sum or mean of squared errors.

## 5. Cost/Loss Functions and Gradient Descent

A loss function measures how wrong an individual prediction is, while a cost or objective function often summarizes loss across the training dataset. Terminology varies across libraries and textbooks.

**Mean Squared Error (MSE):** average of squared differences between actual and predicted values. Squaring prevents positive and negative errors from canceling and gives larger errors more influence.

**Gradient descent:** an optimization algorithm that repeatedly updates model parameters in the direction that reduces the objective function.

**Learning rate:** controls the size of each parameter update. Too large may overshoot the minimum; too small may make training unnecessarily slow.

**Important:** gradient descent is one optimization method. Ordinary least squares linear regression can also be solved analytically.

## 6. Logistic Regression

Despite its name, logistic regression is primarily a **classification** algorithm. It begins with a linear score such as  $z = w \cdot x + b$  and transforms that score into a probability.

**Binary logistic regression:** uses the sigmoid function to map any real-valued score to a value between 0 and 1.

The output can be interpreted as an estimated probability for the positive class, subject to the model assumptions and calibration quality.

**Decision boundary:** after selecting a probability threshold, observations on different sides of the threshold are assigned to different classes. In feature space, the resulting boundary is linear for standard logistic regression unless nonlinear features are introduced.

## 7. Sigmoid Function and Classification Threshold

**Sigmoid:**  $\sigma(z) = 1 / (1 + e^{-z})$ . Its output lies between 0 and 1.

Example: if the model predicts 0.82 and the threshold is 0.50, the observation is classified as the positive class. If the prediction is 0.31, it is classified as the negative class.

**The threshold does not have to be 0.50.** It should be selected according to the business objective and the relative cost of false positives and false negatives.

For high-risk applications, selecting a threshold randomly is poor practice. Threshold selection should use validation data and metrics such as precision, recall, F1 score, ROC analysis, PR curves, expected cost, or domain-specific constraints.

## 8. Softmax and Multiclass Classification

Softmax converts a vector of class scores (logits) into a probability distribution. Each probability is between 0 and 1, and all class probabilities sum to 1.

Example: Cat = 0.10, Dog = 0.25, Monkey = 0.55, Bird = 0.10. The highest probability is Monkey, so an argmax decision predicts Monkey.

**Important distinction:** sigmoid is commonly used for binary classification and also for independent labels in multilabel classification. Softmax is commonly used when classes are mutually exclusive in a multiclass problem.

## 9. Confusion Matrix

A confusion matrix compares actual labels with predicted labels for a classification model.

**True Positive (TP):** actual positive, predicted positive.

**True Negative (TN):** actual negative, predicted negative.

**False Positive (FP):** actual negative, predicted positive.

**False Negative (FN):** actual positive, predicted negative.

Which error matters more depends on the application. In medical screening, missing a genuinely positive case may be especially costly. In other applications, false alarms may carry the greater cost.

## 10. Classification Metrics

**Accuracy:**  $(TP + TN) / \text{Total}$ . Useful when classes are reasonably balanced and error costs are similar.

**Precision:**  $TP / (TP + FP)$ . Of all predicted positives, how many were actually positive?

**Recall (Sensitivity):**  $TP / (TP + FN)$ . Of all actual positives, how many did the model find?

**Specificity:**  $TN / (TN + FP)$ . Of all actual negatives, how many were correctly identified?

**F1 score:** harmonic mean of precision and recall. Useful when balancing both is important.

**Critical point:** high accuracy can be misleading on imbalanced data. A model predicting the majority class every time may achieve high accuracy while failing at the real task.

## 11. ROC Curve and AUC

The ROC curve evaluates a binary classifier across many thresholds by plotting True Positive Rate (recall) against False Positive Rate.

**False Positive Rate:**  $FP / (FP + TN)$ .

**AUC:** Area Under the ROC Curve. A higher AUC generally indicates better ranking ability across thresholds. Roughly, it measures how well the model tends to rank a randomly chosen positive example above a

randomly chosen negative example.

**Important:** ROC-AUC does not automatically provide the best operating threshold. Threshold choice still depends on business costs and validation objectives. For highly imbalanced problems, the Precision–Recall curve may be more informative.

## 12. Decision Trees

A decision tree repeatedly splits data using feature-based rules. Each internal node represents a decision, each branch represents an outcome of that decision, and each leaf produces a prediction.

For classification, common split criteria include Gini impurity and entropy/information gain. For regression, criteria often measure reduction in squared error or variance.

**Advantages:** interpretable, handles nonlinear relationships, little feature scaling required.

**Disadvantages:** a deep tree can overfit and can be unstable because small data changes may create a different tree.

## 13. Random Forest

A random forest is an ensemble of many decision trees. Each tree is typically trained using a bootstrap sample of the data, and random subsets of features are considered at splits.

For classification, predictions are commonly combined by voting or averaged class probabilities. For regression, tree predictions are averaged.

**Why it helps:** combining many decorrelated trees generally reduces variance and improves generalization compared with one unrestricted decision tree.

**Trade-off:** it is less interpretable and more computationally expensive than a single tree.

## 14. Boosting

Boosting builds models sequentially. Each new learner focuses on correcting errors or residuals left by previous learners.

Common families include AdaBoost, Gradient Boosting, XGBoost, LightGBM, and CatBoost.

**Bagging vs boosting:** random forests primarily use bagging and parallel diversity; boosting is sequential and focuses progressively on difficult errors.

Boosting can be extremely powerful on structured/tabular data, but poor hyperparameter choices can lead to overfitting.

## 15. Overfitting and Underfitting

**Underfitting:** the model is too simple to capture the underlying pattern. Performance is poor on both training and unseen data.

**Overfitting:** the model learns the training data too specifically, including noise, and performs much worse on unseen data.

**Good generalization:** the goal is not maximum training accuracy. The goal is strong performance on new, unseen data.

Common controls include train/validation/test splitting, cross-validation, regularization, pruning, early stopping, feature selection, more representative data, and appropriate model complexity.

## 16. Training, Validation, and Testing

**Training set:** used to fit model parameters.

**Validation set:** used to select hyperparameters, thresholds, and model configurations.

**Test set:** reserved for final unbiased evaluation after model-development decisions are complete.

Repeatedly tuning decisions based on the test set causes information leakage. In practice, cross-validation is often used when data is limited.

## 17. End-to-End Machine Learning Workflow

1. Define the business problem and target.
2. Collect and understand the data.
3. Clean missing, invalid, duplicate, or inconsistent values.
4. Perform exploratory data analysis.
5. Engineer/select useful features.
6. Split the data correctly.
7. Build a baseline model.
8. Train candidate models.
9. Evaluate using metrics appropriate to the problem.
10. Tune hyperparameters and decision thresholds using validation data.
11. Perform final test evaluation.
12. Deploy, monitor drift/performance, and retrain when justified.

## 18. Quick Model Selection Guide

**Predicting a number?** Start with regression models.

**Predicting one of two classes?** Binary classification.

**Choosing one class among many?** Multiclass classification.

**Assigning several labels simultaneously?** Multilabel classification.

**No target labels and looking for groups?** Clustering/unsupervised learning.

**Sequential decisions with rewards?** Reinforcement learning.

## 19. Common Misconceptions Corrected

- Machine learning does not literally make a machine think exactly like a human.
- Unsupervised learning does not normally perform supervised regression/classification without labels; it discovers structure such as clusters or latent representations.
- Multiclass and multilabel classification are different.
- Logistic regression does not simply apply sigmoid directly to  $y = mx + c$  and magically create classes; it models class probability through a linear predictor and logistic link.

- A probability threshold should not be selected randomly for serious applications.
- ROC-AUC measures ranking performance across thresholds; it does not by itself determine the operational threshold.
- A best-fit regression line is selected by optimizing a loss/objective, not by visually passing through the most points.

## 20. Interview-Ready Summary

**What is machine learning?** Machine learning is a field in which algorithms learn patterns from data so they can make predictions or decisions on new data without manually coding every rule.

**Supervised vs unsupervised?** Supervised learning uses labeled targets; unsupervised learning discovers structure in unlabeled data.

**Regression vs classification?** Regression predicts continuous values; classification predicts categories.

**Why logistic regression for classification?** It models a linear score and transforms it into a probability, commonly using the sigmoid function for binary classification.

**Precision vs recall?** Precision asks how many predicted positives were correct; recall asks how many actual positives were found.

**Why random forest?** It combines many randomized decision trees to reduce variance and improve generalization.

**What is overfitting?** Strong training performance but poor performance on unseen data because the model learned training-specific noise or details.

# Final Revision Map

| Problem                      | Typical Approach          | Example                   |
|------------------------------|---------------------------|---------------------------|
| Continuous target            | Regression                | House price               |
| Two classes                  | Binary classification     | Fraud / Not fraud         |
| One of many classes          | Multiclass classification | Cat / Dog / Bird          |
| Several simultaneous labels  | Multilabel classification | Image contains Cat + Dog  |
| No target labels             | Unsupervised learning     | Customer segmentation     |
| Sequential actions + rewards | Reinforcement learning    | Control / policy learning |

**Study order:** ML basics → learning types → regression/classification → linear regression → loss and optimization → logistic regression → sigmoid/softmax → confusion matrix and metrics → ROC/AUC → trees → random forest → boosting → overfitting and model evaluation.